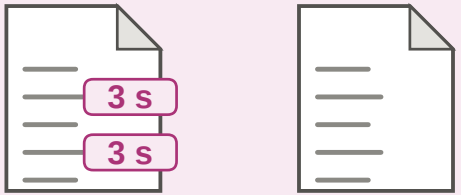
Do speech benchmarks measure cognition, or how the speech was transcribed?

1 Two ways to transcribe



pauses written | **no pauses**

sessions without annotations

HC	**48%**
MCI	**18%**
Dem.	**30%**

2 Text models learn them

← healthy impaired →

removing annotations

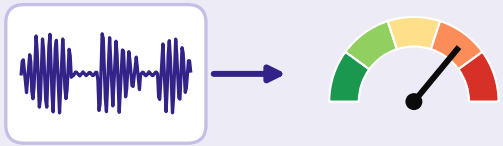
AUC −0.07 to −0.12

timed pauses −0.06 · timer −0.03

only where annotations exist

3 Annotation-free control

ASR + VAD → healthy norms



audio only | risk score

0.81

test AUC [0.70, 0.91]

same under both conventions

Measure pauses from audio · document transcription conventions · report annotation-free controls